

WORKED EXAMPLE

The same wrong prediction, two loss functions

A single sigmoid output predicts $p=0.01$ while the known target is $t=1$.

$$p = \sigma(z) = \frac{1}{1 + e^{-z}}, \quad z = wx + b$$

$$x = 2, \quad b = 0, \quad z = \ln \frac{0.01}{0.99} = -4.59511985$$

$$w = z/2 = -2.29755993$$

The probability is close to zero, so this target-1 prediction is confidently wrong. Sigmoid saturation means the output changes very little when its input z changes a little.

Lecture4 motivates the saturation problem using squared divergence. This is a separate study example using the handout binary target convention t in $\{0,1\}$.

WORKED EXAMPLE

State the loss scaling before differentiating

The lecture PDF page12 uses the unhalved square (y-d)-squared.

$$L_{half} = \frac{1}{2}(p - t)^2, \quad L_{BCE} = -[t \ln p + (1 - t) \ln(1 - p)]$$

$$L_{half} = \frac{1}{2}(0.01 - 1)^2 = 0.49005$$

$$L_{BCE} = -\ln(0.01) = 4.60517019$$

Our half-square convention removes the factor 2 from its derivative. For the lecture unhalved square, double this loss and its gradients: loss 0.9801.

These loss values use different scales and definitions. A smaller numerical loss across two different formulas does not by itself mean a better prediction. Compare their gradient responses at the same prediction.

WORKED EXAMPLE

Half-squared error keeps the small sigmoid slope

Use the chain rule through p before reaching z .

$$\frac{dL_{half}}{dp} = p - t = 0.01 - 1 = -0.99$$

$$\frac{dp}{dz} = p(1 - p) = 0.01(0.99) = 0.0099$$

$$\frac{dL_{half}}{dz} = \frac{dL_{half}}{dp} \frac{dp}{dz}$$

$$= (-0.99)(0.0099) = -0.009801$$

The prediction is badly wrong, yet the derivative with respect to z is small because the sigmoid slope is small. For the unhalved square, the derivative is twice this: -0.019602.

WORKED EXAMPLE

BCE cancels the sigmoid factor

This simplification is specific to sigmoid output with binary cross-entropy.

$$\frac{dL_{BCE}}{dp} = -\frac{t}{p} + \frac{1-t}{1-p}$$

$$\frac{dL_{BCE}}{dz} = \left(-\frac{t}{p} + \frac{1-t}{1-p}\right)p(1-p)$$

$$= -t(1-p) + (1-t)p = p - t$$

$$t = 1: \quad (-1/0.01)(0.0099) = (-100)(0.0099) = -0.99$$

Cancellation assumes $0 < p < 1$, as holds for finite sigmoid logits in exact arithmetic. Do not multiply the simplified $p-t$ by another sigmoid slope.

WORKED EXAMPLE

Translate each output derivative into parameters

Let $\delta = dL/dz$. The weight source is $x=2$; the bias source is 1.

$$\frac{dL}{dw} = \frac{dL}{dz} \frac{dz}{dw} = \delta x, \quad \frac{dL}{db} = \delta$$

half: $g_w = (-0.009801)(2) = -0.019602, \quad g_b = -0.009801$

BCE: $g_w = (-0.99)(2) = -1.98, \quad g_b = -0.99$

$$\frac{|\delta_{BCE}|}{|\delta_{half}|} = \frac{0.99}{0.009801} \approx 101.01$$

Subtracting either gradient increases z and the target-1 probability. Their magnitudes differ because the two losses weight this error differently.

WORKED EXAMPLE

one step with half-squared error

Update both w and b from the same original state, using $\eta=0.1$.

$$w_{new} = -2.29755993 - 0.1(-0.01960200) = -2.29559973$$

$$b_{new} = 0 - 0.1(-0.00980100) = 0.00098010$$

$$z_{new} = 2(-2.29559973) + (0.00098010) = -4.59021935$$

$$p_{new} = \frac{1}{1 + e^{-(-4.59021935)}} = 0.01004863$$

The updated prediction uses both changed parameters. The comparison starts from the same original weight and bias for each loss; these are alternative steps, not consecutive steps.

WORKED EXAMPLE

one step with BCE

Update both w and b from the same original state, using $\eta=0.1$.

$$w_{new} = -2.29755993 - 0.1(-1.98000000) = -2.09955993$$

$$b_{new} = 0 - 0.1(-0.99000000) = 0.09900000$$

$$z_{new} = 2(-2.09955993) + (0.09900000) = -4.10011985$$

$$p_{new} = \frac{1}{1 + e^{-(-4.10011985)}} = 0.01630058$$

The updated prediction uses both changed parameters. The comparison starts from the same original weight and bias for each loss; these are alternative steps, not consecutive steps.

WORKED EXAMPLE

Why the logit change contains x-squared

Combine the updates only after you understand both parameter contributions.

$$\begin{aligned} z_{new} &= (w - \eta\delta x)x + (b - \eta\delta) \\ &= wx + b - \eta\delta(x^2 + 1) = z - \eta\delta(x^2 + 1) \end{aligned}$$

$$x = 2: \quad x^2 + 1 = 5$$

$$\text{half:} \quad \Delta z = -0.1(-0.009801)(5) = 0.0049005$$

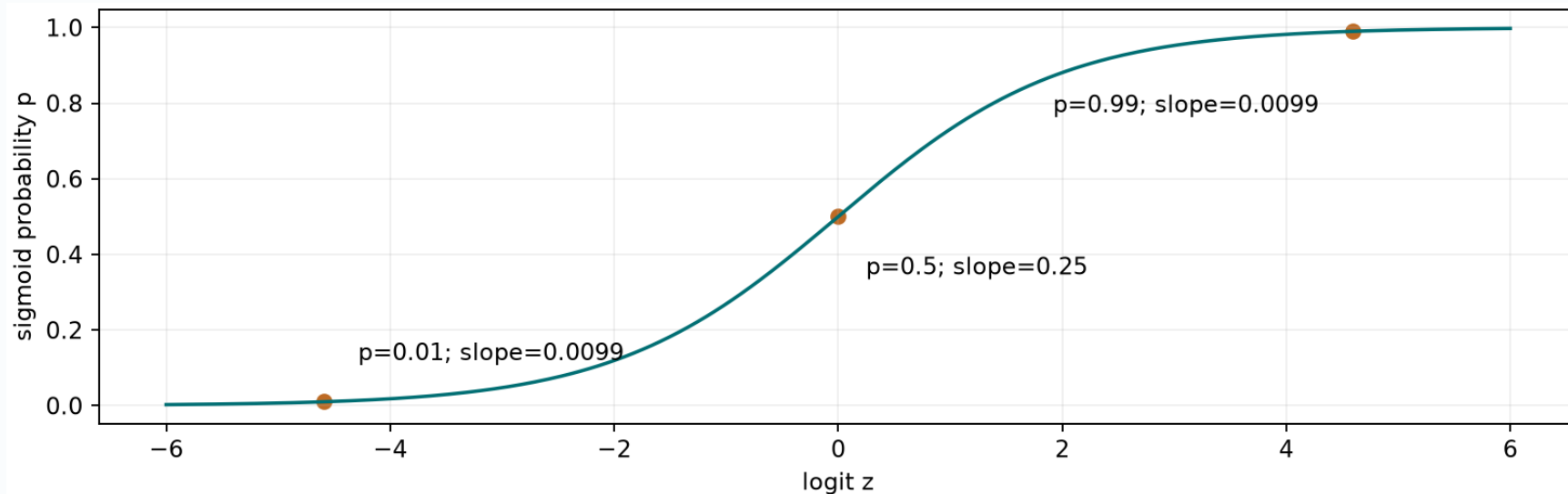
$$\text{BCE:} \quad \Delta z = -0.1(-0.99)(5) = 0.495$$

This is a same-example logit change after updating both its weight and bias. A bias-only update would omit the x-squared term.

WORKED EXAMPLE

Where sigmoid becomes insensitive

The same sigmoid is used with either loss.



The slope is 0.0099 at $p=0.01$ or 0.99 , versus 0.25 at $p=0.5$. BCE removes this output factor from the logit derivative; it does not remove every hidden-layer sigmoid factor or guarantee easy optimization.

INDEPENDENT PRACTICE

Your turn: a wrong target-zero prediction

Use a new probability and recompute all quantities.

$$p = 0.9, \quad t = 0, \quad x = 2, \quad b = 0$$

$$z = \ln(9), \quad w = \ln(9)/2, \quad \eta = 0.1$$

Find both losses, the sigmoid slope, both derivatives with respect to p and z , and the weight/bias gradients. Then make one simultaneous w, b update for each loss and recompute the probability.

Explain why the new logit decreases. State what would change if the squared loss had no factor $1/2$.

WORKED ANSWER

Practice: losses and full chain-rule arithmetic

The target is zero, so BCE uses the logarithm of 1-p.

$$L_{half} = \frac{1}{2}(0.9 - 0)^2 = 0.405, \quad L_{BCE} = -\ln(0.1) = 2.30258509$$

$$\frac{dp}{dz} = 0.9(1 - 0.9) = 0.09$$

$$\text{half: } \frac{dL}{dp} = 0.9, \quad \delta = 0.9(0.09) = 0.081$$

$$\text{BCE: } \frac{dL}{dp} = 1/(1 - 0.9) = 10, \quad \delta = 10(0.09) = 0.9$$

$$\text{half: } (g_w, g_b) = (0.162, 0.081); \quad \text{BCE: } (1.8, 0.9)$$

Positive parameter gradients are subtracted, reducing w and b. For this positive input, both changes reduce z and therefore reduce the wrong high probability.

WORKED ANSWER

Practice: one step with half-squared error

Update both w and b from the same original state, using $\eta=0.1$.

$$w_{new} = 1.09861229 - 0.1(0.16200000) = 1.08241229$$

$$b_{new} = 0 - 0.1(0.08100000) = -0.00810000$$

$$z_{new} = 2(1.08241229) + (-0.00810000) = 2.15672458$$

$$p_{new} = \frac{1}{1 + e^{-(2.15672458)}} = 0.89629549$$

The updated prediction uses both changed parameters. The comparison starts from the same original weight and bias for each loss; these are alternative steps, not consecutive steps.

WORKED ANSWER

Practice: one step with BCE

Update both w and b from the same original state, using $\eta=0.1$.

$$w_{new} = 1.09861229 - 0.1(1.80000000) = 0.91861229$$

$$b_{new} = 0 - 0.1(0.90000000) = -0.09000000$$

$$z_{new} = 2(0.91861229) + (-0.09000000) = 1.74722458$$

$$p_{new} = \frac{1}{1 + e^{-(1.74722458)}} = 0.85160240$$

The updated prediction uses both changed parameters. The comparison starts from the same original weight and bias for each loss; these are alternative steps, not consecutive steps.

WORKED ANSWER

Practice checks and the scope of the conclusion

Same prediction and architecture; different loss derivatives.

half: $z_{new} = \ln 9 - 0.1(0.081)(5) = \ln 9 - 0.0405$

BCE: $z_{new} = \ln 9 - 0.1(0.9)(5) = \ln 9 - 0.45$

Using an unhalved square doubles its loss and gradients. Keeping the same learning rate would therefore double that parameter displacement.

Do not threshold the probability before the loss, add an extra sigmoid derivative to p-t, confuse a loss derivative with a weight derivative, or infer that BCE eliminates hidden-layer vanishing gradients.